

Weirui Wang

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Fall 2027 Ph.D. Applicant · Character Animation, Motion Generation, Rigging, and Character Control

🎓 EDUCATION

ShanghaiTech University, Shanghai, China

Sep. 2024 – Jun. 2027 (Expected)

M.S. in Computer Science

Advisors: Prof. Jingyi Yu and Prof. Lan Xu

Shandong University, Jinan, China

Sep. 2020 – Jun. 2024

B.Eng. in Control Science and Engineering

Overall Average: 91.57/100

Selected Coursework: Automatic Control, Robotics, Robot Systems, and Intelligent Sensing

📖 PUBLICATIONS

- **ACT: A Unified Framework for Rigging and Animating Characters with Arbitrary Topologies**

ACM TOG (SIGGRAPH 2026)

Pengyu Long*, Weirui Wang*, Qingcheng Zhao, Xiaoyang Guo, Xiaoyu Pan, Qixuan Zhang, Jiaqing Zhou, Tianlei Hu, Wei Yang, Lan Xu, Jingyi Yu (* Equal contribution)

🔬 RESEARCH EXPERIENCE

ShanghaiTech University, Shanghai, China

Sep. 2024 – Present

Master's Student Researcher, advised by Prof. Jingyi Yu and Prof. Lan Xu

ACT: A Unified Framework for Rigging and Animating Characters with Arbitrary Topologies

ACM TOG / SIGGRAPH 2026, Co-first Author

- Observed that rigging and motion are mutually dependent: a rig should be designed for the motions it must support, while generated motion must respect the character's skeleton and body structure. This motivated ACT to learn rigging and animation together rather than treating them as separate steps.
- **Developed the motion branch, built the motion dataset, and proposed ACT's unified structure-motion formulation.** Implemented the core model and training pipeline to generate editable skeletons, skinning weights, and controllable, text-driven motions for characters with diverse body structures. ACT reduces skeleton errors across three metrics by **45–49%** and motion FID by **48%** over the strongest baselines.

AniRig: Recovering Animation-Ready Rigged Assets from Arbitrary-Topology 3D Characters

First-Author Research Project

- Proposed a method that converts a single posed 3D character into a complete animation-ready asset, including a neutral rest-pose mesh, skeleton, and skinning weights.
- Designed the method to handle folded limbs, touching body parts, and complex character shapes. The recovered assets support re-posing and animation reuse, connecting 3D asset creation with downstream character animation. **Led the project end to end.**

🏢 INDUSTRY EXPERIENCE

Deemos Technologies Inc., Shanghai, China

Apr. 2024 – Present

Research Intern

- **Character Rigging:** Adapted ACT to production 3D assets with broken surfaces and irregular structures, improving robust skeleton and skinning prediction for motion-ready characters. Deployed in Hyper3D and ByteDance workflows.
- **Facial Enhancement:** Built a reference-guided image generation pipeline that improves identity consistency and facial detail while preserving the character's 3D shape. Deployed in Hyper3D Rodin.

⚙️ RESEARCH SKILLS

Character Animation: Character Rigging, Skinning, Text-Driven Motion Generation, Motion In-betweening, Forward and Inverse Kinematics

Learning Methods and Tools: Diffusion Models, Transformers, 3D Representation Learning, Python, PyTorch, Blender, L^AT_EX